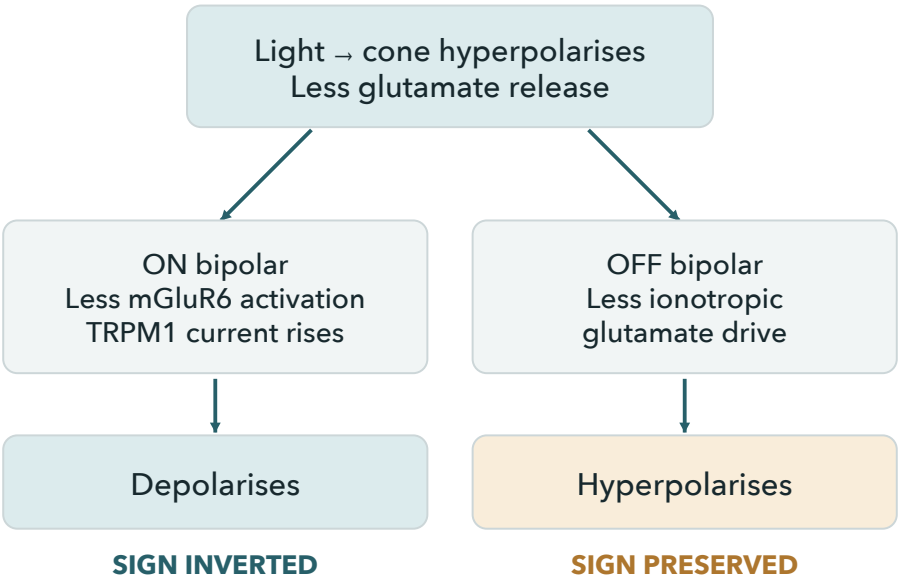


Retinal contrast circuits

A light increment has opposite signs in the ON and OFF pathways.

ON and OFF: invert or preserve

A | One cone signal, two bipolar responses



Photoreceptors and bipolar cells mainly use graded potentials. Most ganglion cells encode output as action potentials.

Cone bipolar response

State	ON bipolar	OFF bipolar
Dark: more glutamate	mGluR6 signalling suppresses TRPM1 current	Ionotropic receptors provide depolarising drive
Light: less glutamate	Released from inhibition; depolarises	Less depolarising drive; hyperpolarises
mGluR6 kept active	Light-evoked ON response reduced	Separate OFF receptor pathway can remain responsive

Fast check

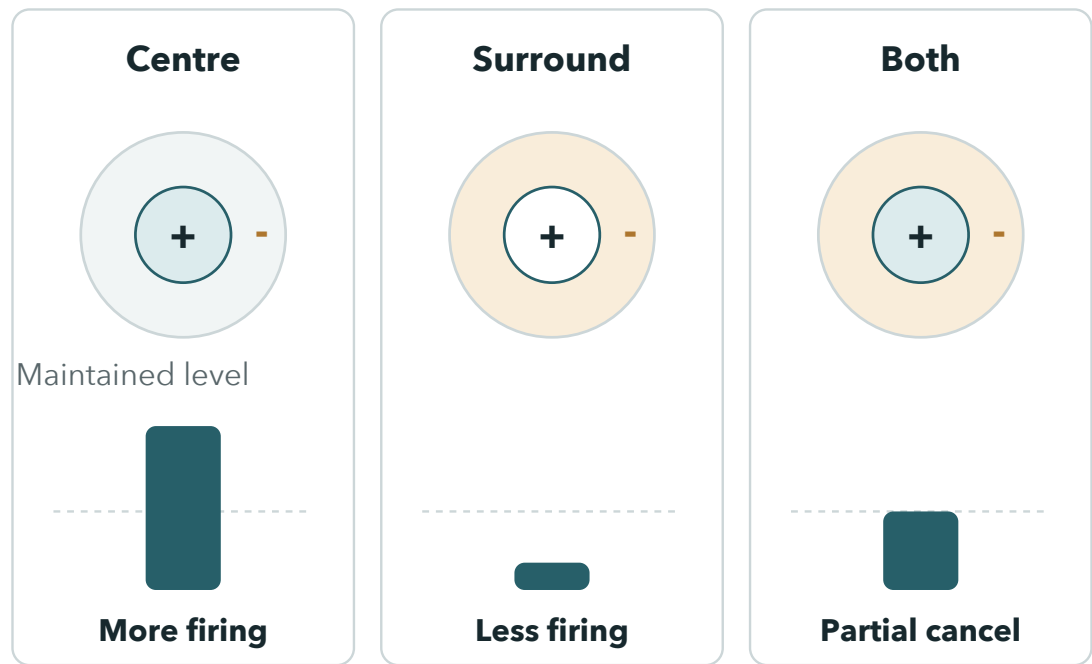
“ON” means preference for a light increment, not that the bipolar cell releases the same response sign as its photoreceptor.

Figure reference

A: Webvision bipolar pathways; supplied TBL2 retinal processing. Original schematic.

Centre-surround: encode a difference

B | ON-centre ganglion responses



Tinted centre or annulus shows the illuminated region. Responses are relative to maintained firing; OFF-centre cells have opposite preference.

Where is the comparison built?

Circuit	Main role
Horizontal cells: outer plexiform layer	Lateral interactions contribute to antagonistic surrounds
Amacrine cells: inner plexiform layer	Shape timing and other ganglion-cell computations
Ganglion receptive field	Centre and surround together emphasise local differences

Fast check

A bright centre can increase firing; adding a bright surround can reduce that increase. A large uniform patch often drives less contrast signal than a centre-selective spot.

Figure reference

B: Webvision ganglion-cell responses. Original stimulus-response schematic.